Stackelberg Model

Outlines

- Quantity Leadership or Stackelberg Model
- Assumptions
- Leader’s Decision
- Follower’s Decisions
- Market Equilibrium
- Profit Maximization:
- Example

What is the Stackelberg Model?

- The Stackelberg model is like a game between two firms in a market. One firm takes the lead (we call it the leader), and the other follows (we call it the follower). They decide how much to produce one after the other, not at the same time.

Stackelberg model

- In the Stackelberg model with two firms, one firm is designated as the leader, while the other firm is the follower. The leader makes its strategic decision first, and then the follower observes this decision and makes its decision accordingly. This sequential decision-making process differentiates the Stackelberg model from other models like Cournot or Bertrand competition, where firms make their decisions simultaneous.

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Quantity Leadership

- The terms "quantity leadership" and "Stackelberg model" are often used interchangeably, as they refer to the same concept in the context of strategic interaction among firms in oligopolistic markets. Both terms describe a scenario in which one firm, the leader, sets its quantity (or output) first, and then other firms, the followers, observe this quantity and adjust their own quantities accordingly.

Assumptions:

- **Two Firms**: The model considers two firms, one acting as the leader and the other as the follower.
- **Homogeneous Product**: Firms produce identical products, implying perfect substitutability for consumers.
- **Sequential Decision Making**: The leader moves first in setting either output or price, followed by the follower.
- **Perfect Information**: Firms have perfect knowledge of market demand, costs, and rivals' strategies.
- **Profit Maximization**: Firms seek to maximize profits and make decisions accordingly.

Leader's Decision:

- **Leader's Advantage**: The leader moves first and sets its output (or price) anticipating the follower's response.
- **Strategic Decision**: The leader selects its quantity ($Q1$) to maximize its profits, considering the follower's reaction function.
- **Example**: Suppose the leader, Firm 1, chooses to produce $Q1$ units.

Follower's Decision

- **Observation**: The follower observes the leader's output and chooses its own output ($Q2$) accordingly.
- **Reaction Function**: Follower's output is determined by a reaction function, where $Q2$ is a function of $Q1$.
- **Example**: Firm 2 decides its quantity ($Q2$) based on Firm 1's output ($Q1$) to maximize its profits.

Market Equilibrium:

- **Total Output**: Market supply is the sum of leader's and follower's outputs: $Q=Q1+Q2$.
- **Price Determination**: Market price (P) is determined by market demand, given total output Q.
- **Example**: Market price adjusts to ensure that demand equals total supply (Q).

Profit Maximization

- **Profit Functions**: Each firm calculates its profit based on market price P and its own output ($Q1$ for the leader, $Q2$ for the follower).
- **Mathematical Formulation**: Profits ($\pi 1$ for the leader, $\pi 2$ for the follower) are given by $\pi 1 = P \cdot Q1 - C(Q1)$ and $\pi 2 = P \cdot Q2 - C(Q2)$, where $C(Q)$ represents total cost.
- **Example**: Firm 1 and Firm 2 calculate their profits based on the market price and their respective outputs.

Conclusion

- The Stackelberg model offers insights into strategic interactions between firms, emphasizing the advantage of moving first. It's a useful tool for analyzing competition dynamics in various industries.
- Mathematically, the model involves solving optimization problems for each firm and finding equilibrium quantities and prices based on market demand and cost functions.
- This framework helps in understanding how sequential decision-making affects market outcomes and firm strategies in oligopolistic markets.

Price Leadership model

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Price Leadership model

- The Price Leadership model is a concept in economics that describes a market situation where one firm, known as the price leader, sets the price for a product or service, and other firms in the industry, known as followers, adjust their prices accordingly. In this model, the price leader's decisions serve as a reference point for other firms, who typically follow the leader's price to maintain market stability and avoid price competition.

Leader’s Decision

- **Leader's Advantage**: The price leader sets the price first based on its assessment of market conditions, cost structure, and strategic considerations.
- **Strategic Decision**: The leader selects the price (*PL*) to maximize its profits, considering the expected responses of followers and market demand.
- **Example**: Firm A, the price leader, decides to set the price of its product at $10 per unit.

Follower’s Decision

- **Observation**: The followers observe the leader's price and decide whether to follow or set their own prices.
- **Price Following**: Followers typically set their prices slightly below the leader's price to attract customers while maintaining a price advantage.
- **Example**: Firm B and Firm C, the followers, decide to set their prices at $9.50 per unit in response to Firm A's price of $10 per unit.

Market Equilibrium:

- **Consumer Behavior**: Consumers respond to prices set by firms, adjusting their purchases based on their preferences and budget constraints.
- **Total Demand**: Market demand is the sum of demands for each firm's product at their respective prices.
- **Example**: If consumers demand 100 units of the product at $10 per unit, and 150 units at $9.50 per unit, total demand is 250 units.

Profit Maximization

- **Profit Functions**: Each firm calculates its profit based on the price and the quantity sold.
- **Mathematical Formulation**: Profits (π) for each firm are given by $\pi=(P-MC)\times Q$, where MC represents marginal cost and Q is the quantity sold

Conclusion

- The Price Leadership model helps us understand how firms make pricing decisions in a market where one firm takes the lead. It's like a strategic game where each firm tries to set its price to maximize profit while considering the actions of other firms.

Comparing Price Leadership vs Quantity Leadership

- **Decision Variable**: Quantity Leadership focuses on output levels, while Price Leadership focuses on price levels.
- **Strategic Considerations**: Quantity Leadership emphasizes quantity decisions and their impact on market supply, while Price Leadership emphasizes price decisions and their effect on market demand.
- **Market Outcome**: Quantity Leadership may lead to quantity adjustments and potential price impacts, while Price Leadership typically leads to price stability and less price competition.

- **Thank You**